Dialogue-based CALL

Dialogue systems, **conversational agents** and **chatbots** for language learning: challenges and opportunities

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Dialogue-based CALL

dialogue systems / conversational agents / chatbots

What?
- Definition
- Disciplinary approaches

Methodology
Research synthesis

Why?
- Rationale for dialogue-based CALL
- Results from effectiveness studies

How?
- Typology
- Technological approaches

Conclusion
Challenges & opportunities

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Tutor Mike (Ron C. Lee – ESLfast.com)

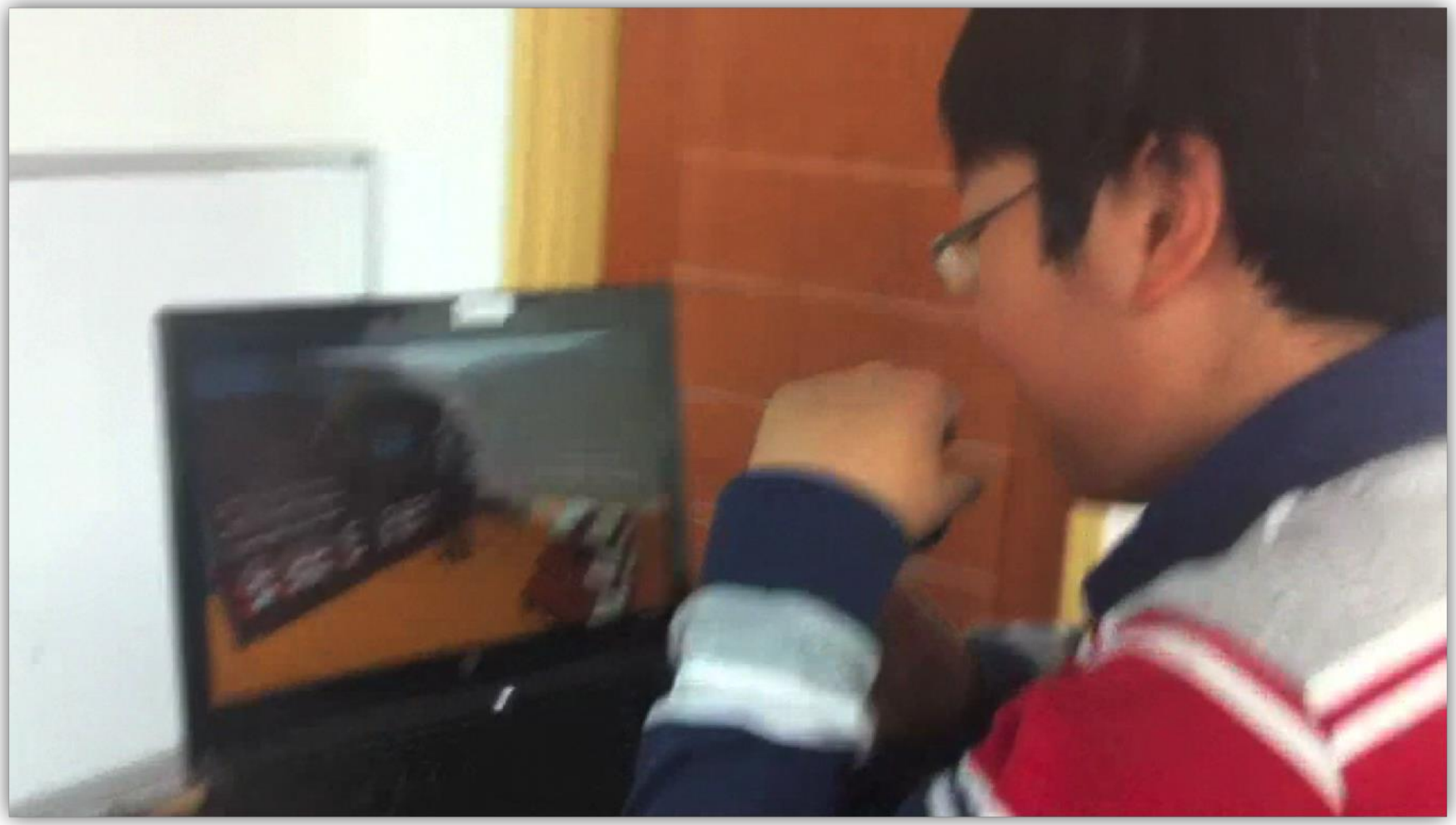
http://www.rong-chang.com/tutor_mike.htm

SPELL system (Morton, Gunson & Jack • University of Edinburgh)

Morton & Jack, 2005, 2010; Morton, Davidson & Jack, 2008; Morton, Gunson & Jack, 2012

POSTECH Immersive English Study (POMY)

(Lee et al. – POSTECH, South Korea)



Tactical Iraqi (Alelo / Johnson et al.)

http://dodsupport.alelo.com/tacticaloperational-language-and-culture-training

Dialogue-based CALL: operational definition

WHAT IT IS

- **dialogue**-based

  synchronous interaction between 2+ speakers, with turn-taking

- interactions **with** a computer

  = Tutorial CALL
  (cf. Heift & Schulze, 2015)

- allowing **negotiation of outcome** through **open learner output**

  (cf. Young, 1988;
  Desmet & Wylin, 2015)

WHAT IT IS NOT

- **item**-based

- interactions **via** a computer

  = CMC in general

- **predetermined outcome** through **closed learner output**

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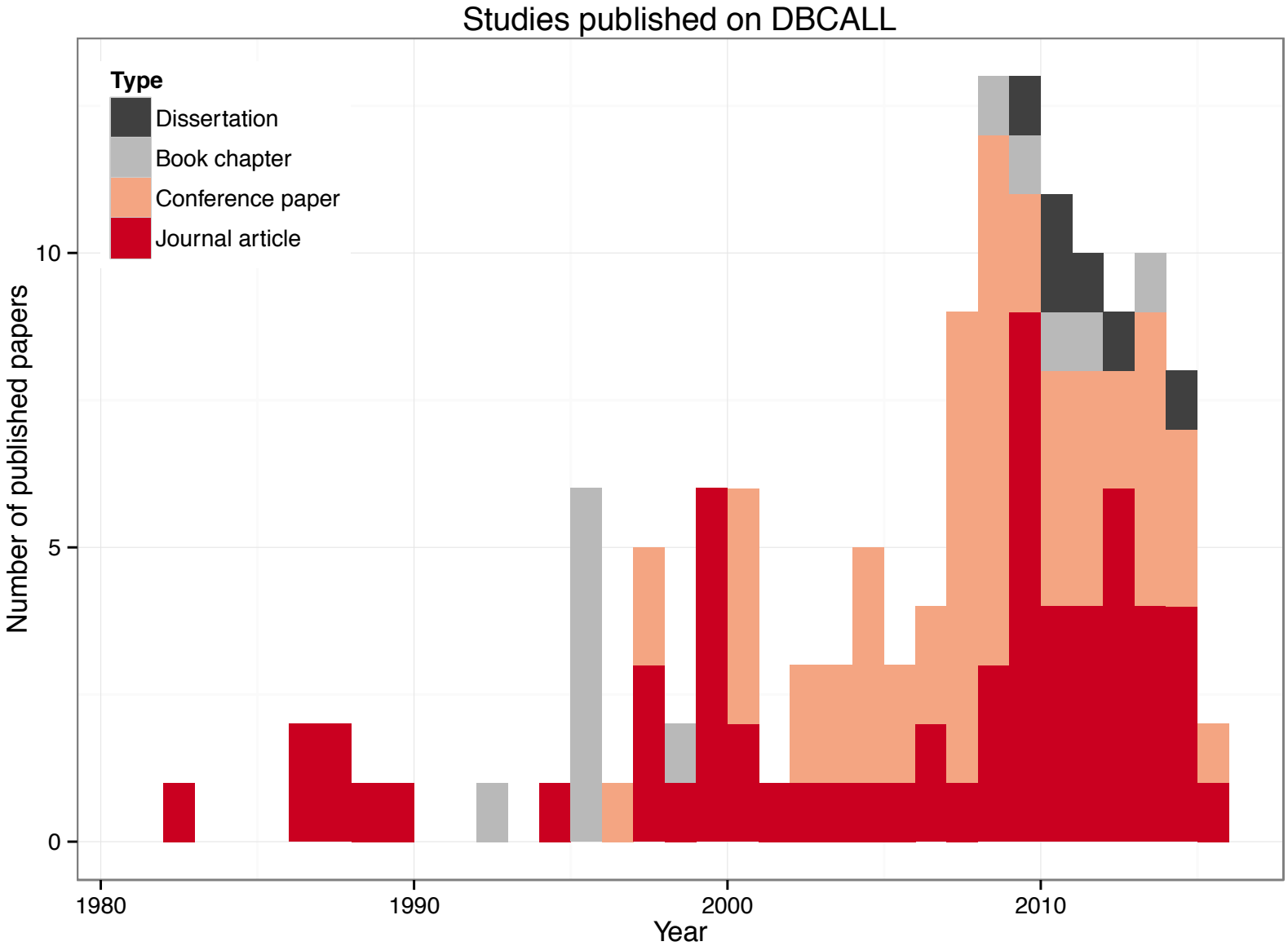
Systematical literature review: Papers collection

- Systematical search on scientific databases:
  - Web of Science, Scopus and ProQuest
  - keywords: chat(ter)bot, conversational agent/companion/system, dialogue system, pedagogical/dialog/virtual agent, dialogue game, human-computer dialogue (…) + language learning/acquisition/teaching, second/foreign language, ESL|EFL, English teaching/learning (…)
  - [results: 604 / 494 / 1003 hits]
- Ancestry (citing) and forward (cited in) search
- Filtering through inclusion & exclusion criteria
- Final corpus:
  139 published peer-reviewed papers
  from 1982 to (April) 2015

Papers collection



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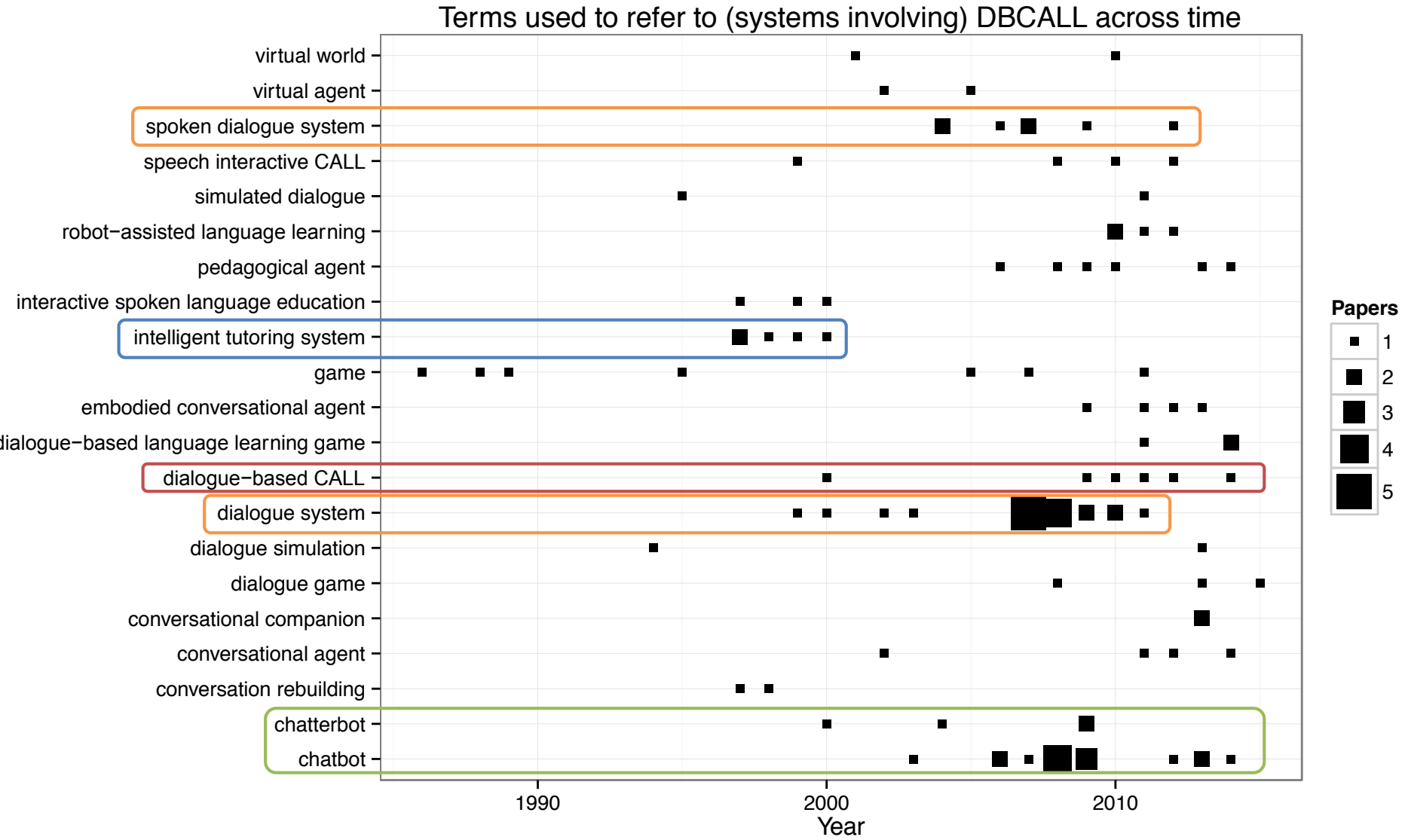
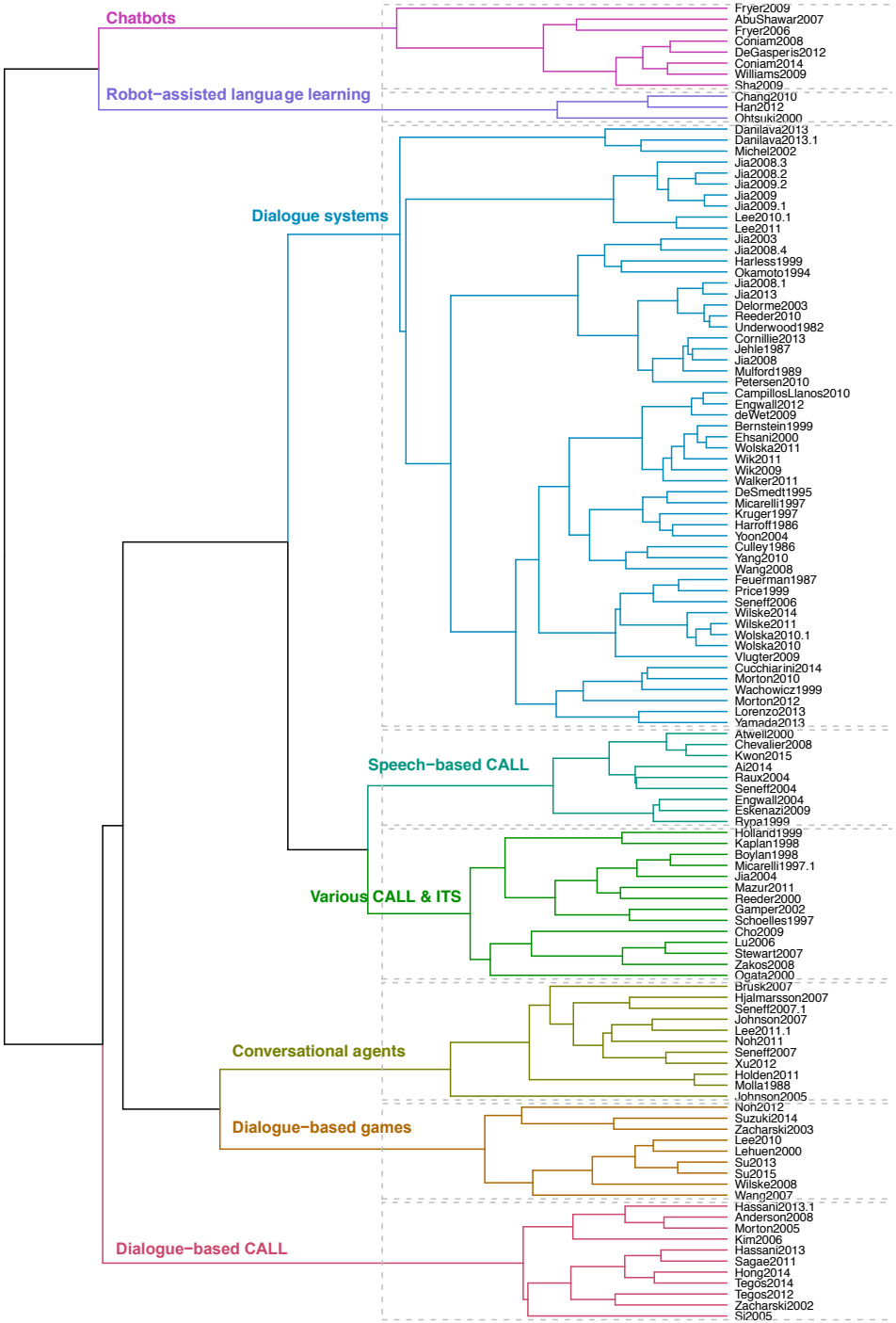
So many different terms…



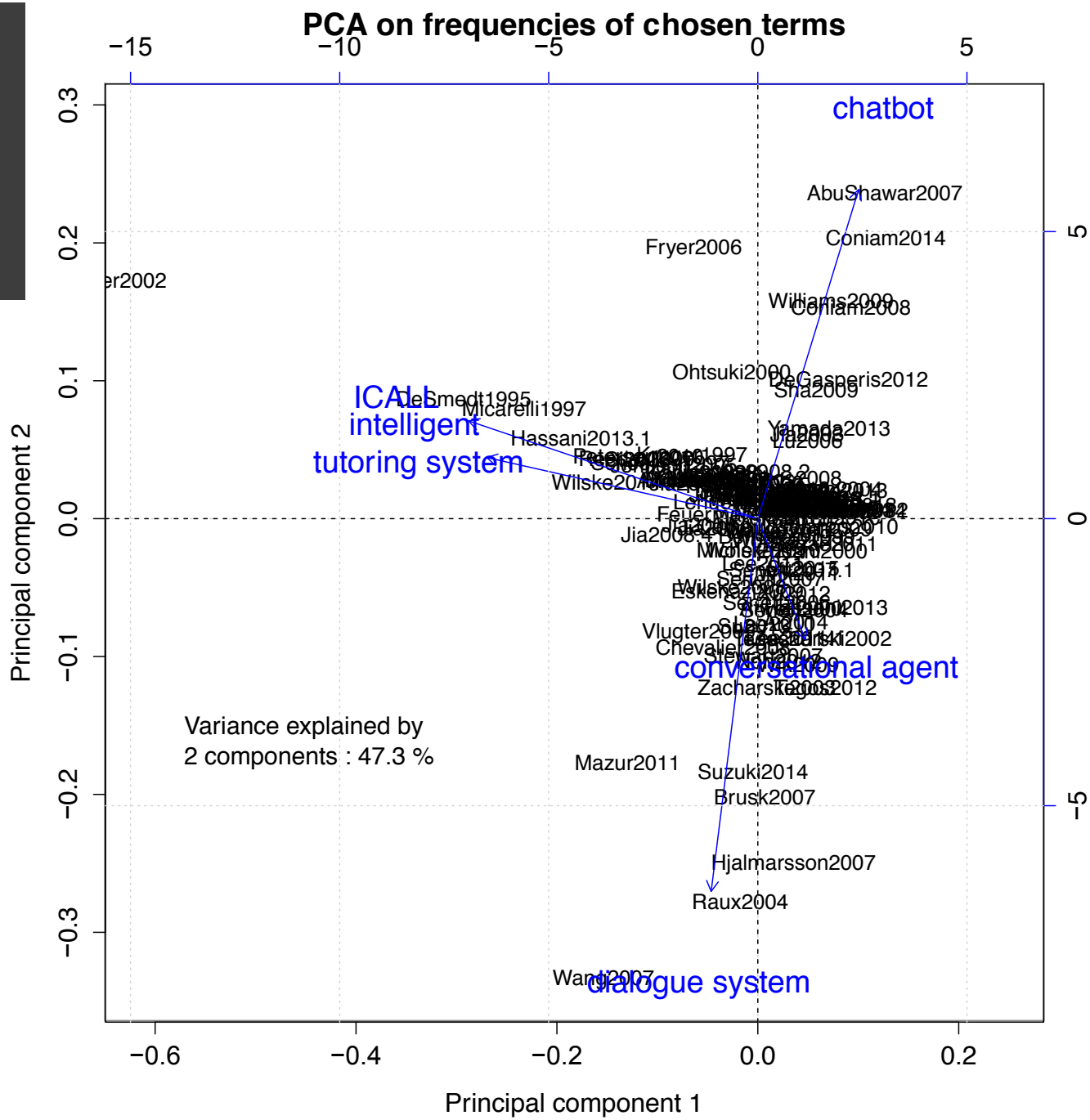
Clusters dendogram of DBCALL papers

Based on accounted frequencies
of selected terms (28) in each paper

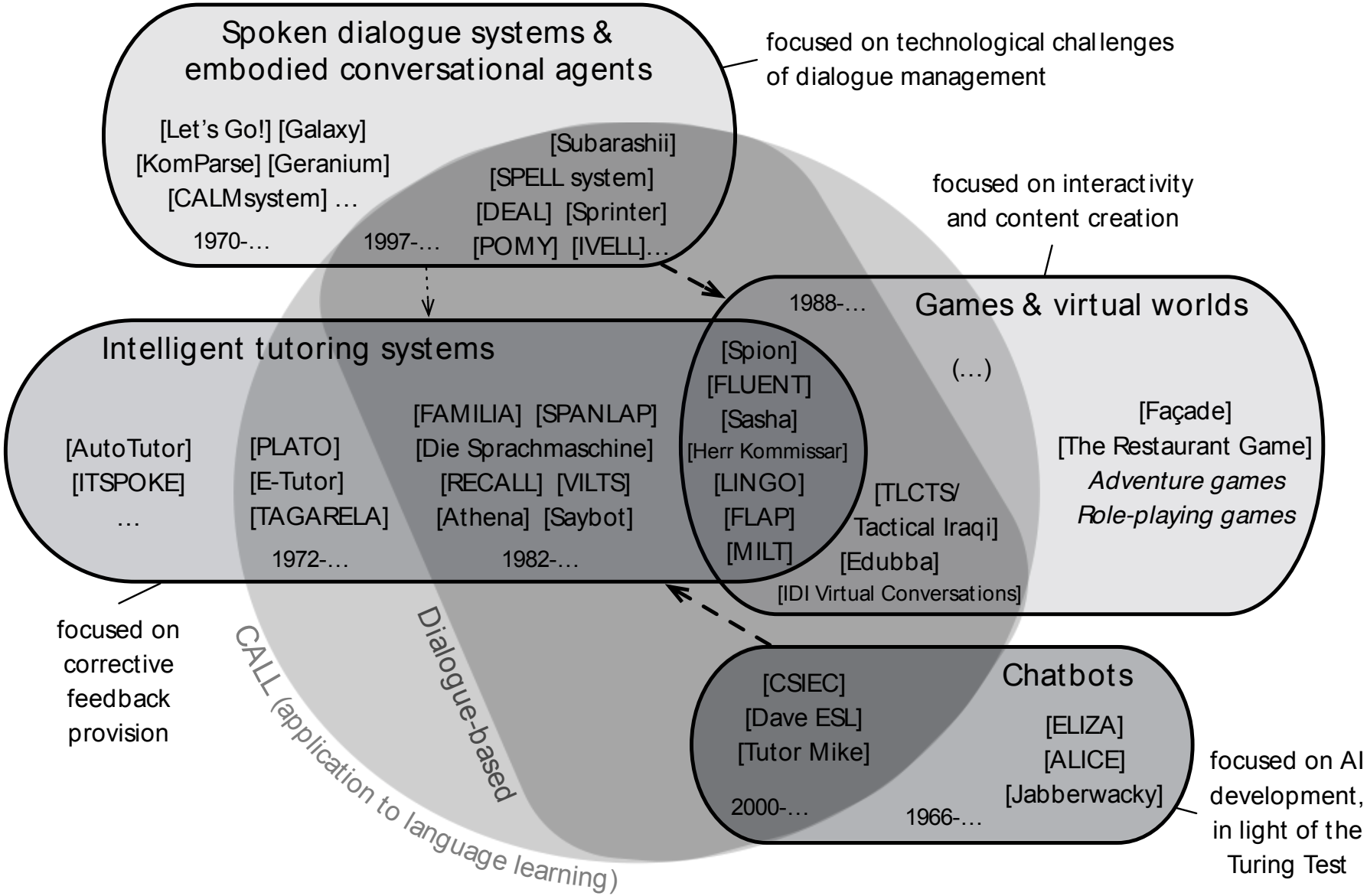


Principal component analysis

Based on accounted frequencies of selected terms (5) in each paper



Disciplinary approaches in DBCALL



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SCMC (chat) ⇒ L2 development

- SCMC: ↗L2 oral proficiency
  (Payne & Whitney, 2002; Payne & Ross, 2005;… Ziegler, 2013; Lin, 2015)
- SCMC: ↗L2 oral proficiency, more than face-to-face
  (Sykes, 2005; Lin, 2015; although not confirmed by Ziegler, 2013)
- Why?
  - **attention** to form (**noticing** and **feedback**)
  - lower anxiety levels (Satar & Özdener, 2008)

DBCALL ⇢ L2 development

- In foreign language teaching contexts: interactions in L2 often very rare

Dialogue-based CALL allows for:

- **Autonomous practice** in a communicative, cultural, authentic and interactive task
  (Wachowicz & Scott, 1999; Fryer & Carpenter, 2006)

  "Virtual immersion" (Ellis & Bogart, 2007; Fryer & Carpenter, 2006)

- Potential impact on **willingness to communicate**

DBCALL: advantages over SCMC

- Available at any moment
- Learner can go at his own pace (repeat, etc.)
- Low anxiety environment (↗ willingness to communicate)
- Potentially fully controllable learning environment (e.g. feedback, learner modeling and adaptivity, motivational support)

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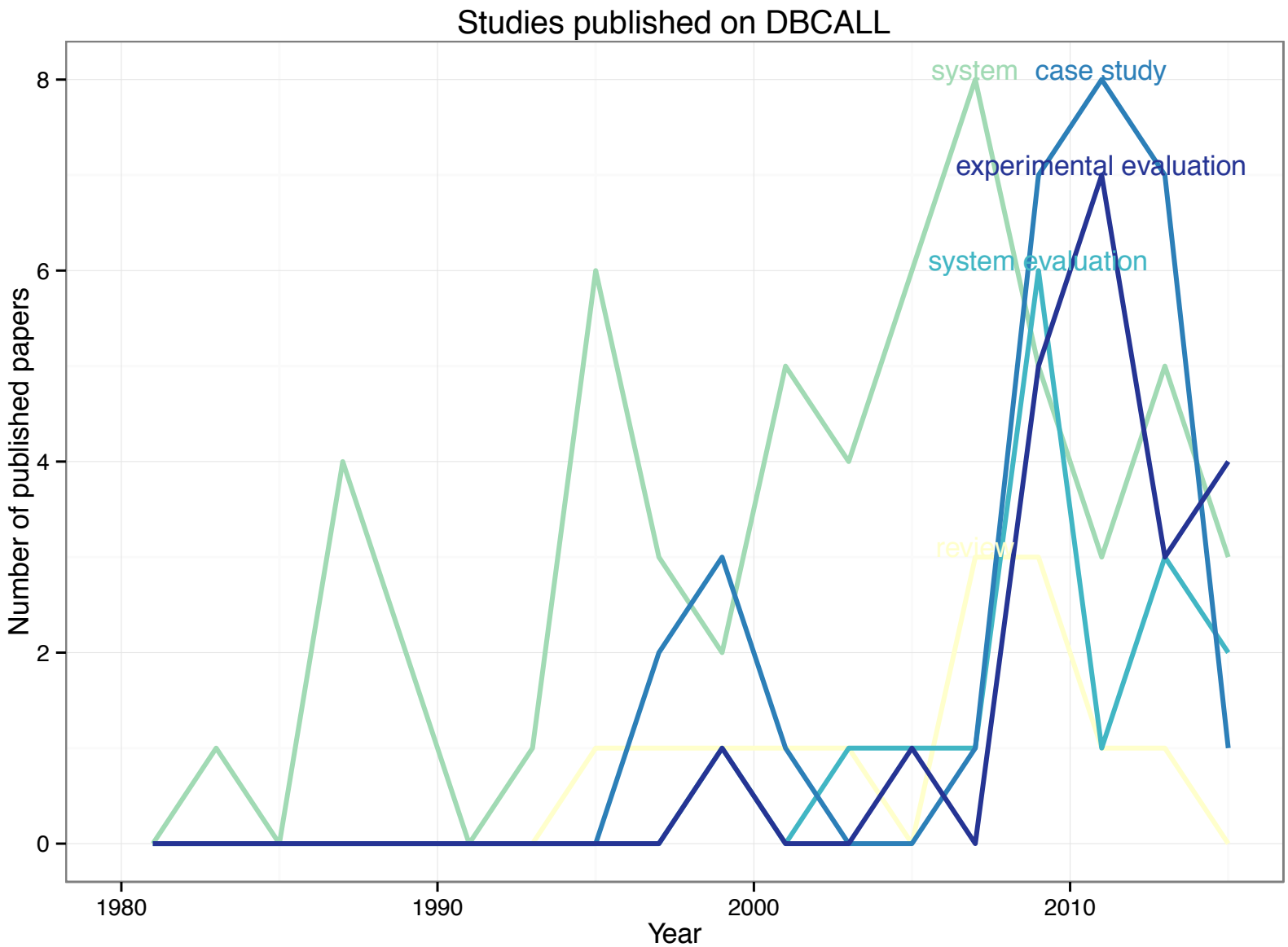
Types of **studies** on dialogue-based CALL

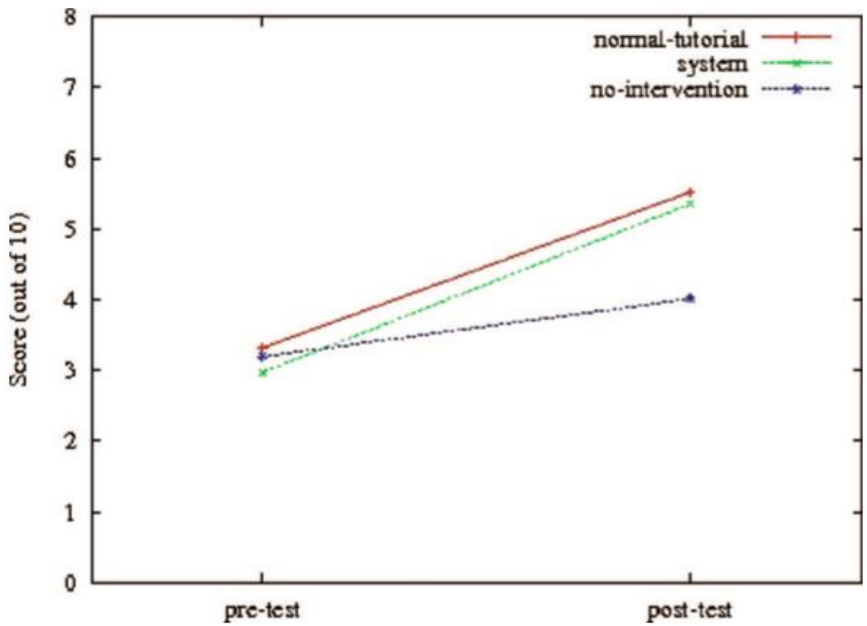


Effectiveness studies on DBCALL

Number of studies by dependent variables and methodological design	**Perceived usefulness**	**Impact on motivation and attitude**	**Impact on language knowledge and skills**
Post	14	1	3
Pre-post		7	22
Pre-post + control		4	12
Pre-post + control with significant results		0	2

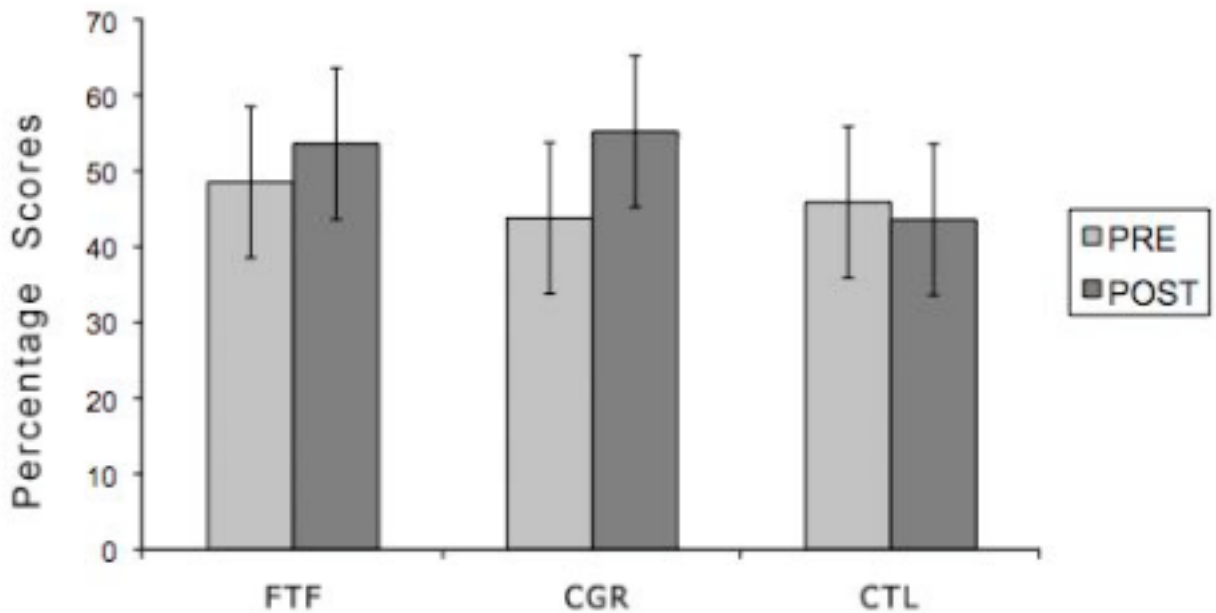
Experimental studies on DBCALL's impact on learning

- Vlugter, Knotta, McDonald & Hall (2009)
  [Te Kaitito]: written DBCALL in Māori
  - compares DBCALL tutorial with in-class tutorial about Māori personal pronoun system



Experimental studies on DBCALL's impact on learning

- Petersen (2010)
  [Sasha]: written DBCALL system in English providing corrective feedback (recast) and answering questions in a "spot the differences" task
  - compares Sasha vs. face-to-face interaction
    => impact on question construction



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Typology of DBCALL systems: modes

	Instruction design	System design
Input mode (learner output)	Written	Text
	Spoken	Audio (+prosodic cues, +gestures, +facial expr.)
Output mode (system output)	Written	Text
	Spoken	Text-to-speech (TTS)
	Spoken + written	Text + TTS
		Embodied agent (gestures, facial expr., gaze, environment…)

Towards a typology of DBCALL systems

Instruction design	Reactive systems	Graph-based systems	Plan-based systems
Systems	• Question-answering systems (Siri…) • Chatbots (ALICE, Tutor Mike…)	Most (spoken) dialogue systems • POMY (Lee et al. 2014) • SPELL system (Morton & Jack, 2010)	• DEAL (Wik, 2011) • (Zacharski, 2002)
Type of activity	**Exercise**	**Activity**	**Task**
Contextualization	Isolated pairs	Abstract situation	Situation + agents' objectives
Negotiation of outcome	Low	Medium	High

Towards a typology of DBCALL systems

System design	Reactive systems	Graph-based systems	Plan-based systems
Systems	• Question-answering systems (Siri…) • Chatbots (ALICE, Tutor Mike…)	Most (spoken) dialogue systems • POMY (Lee et al. 2014) • SPELL system (Morton & Jack, 2010)	• DEAL (Wik, 2011) • (Zacharski, 2002)
Dialogue modeling	**Adjacency pairs**	**Finite-state**	**Dialogue state and planning**
Initiative	User	System (±)	Mixed (±)
Conversational memory	None (amnesia)	State position in the graph	Complex representation of dialogue state

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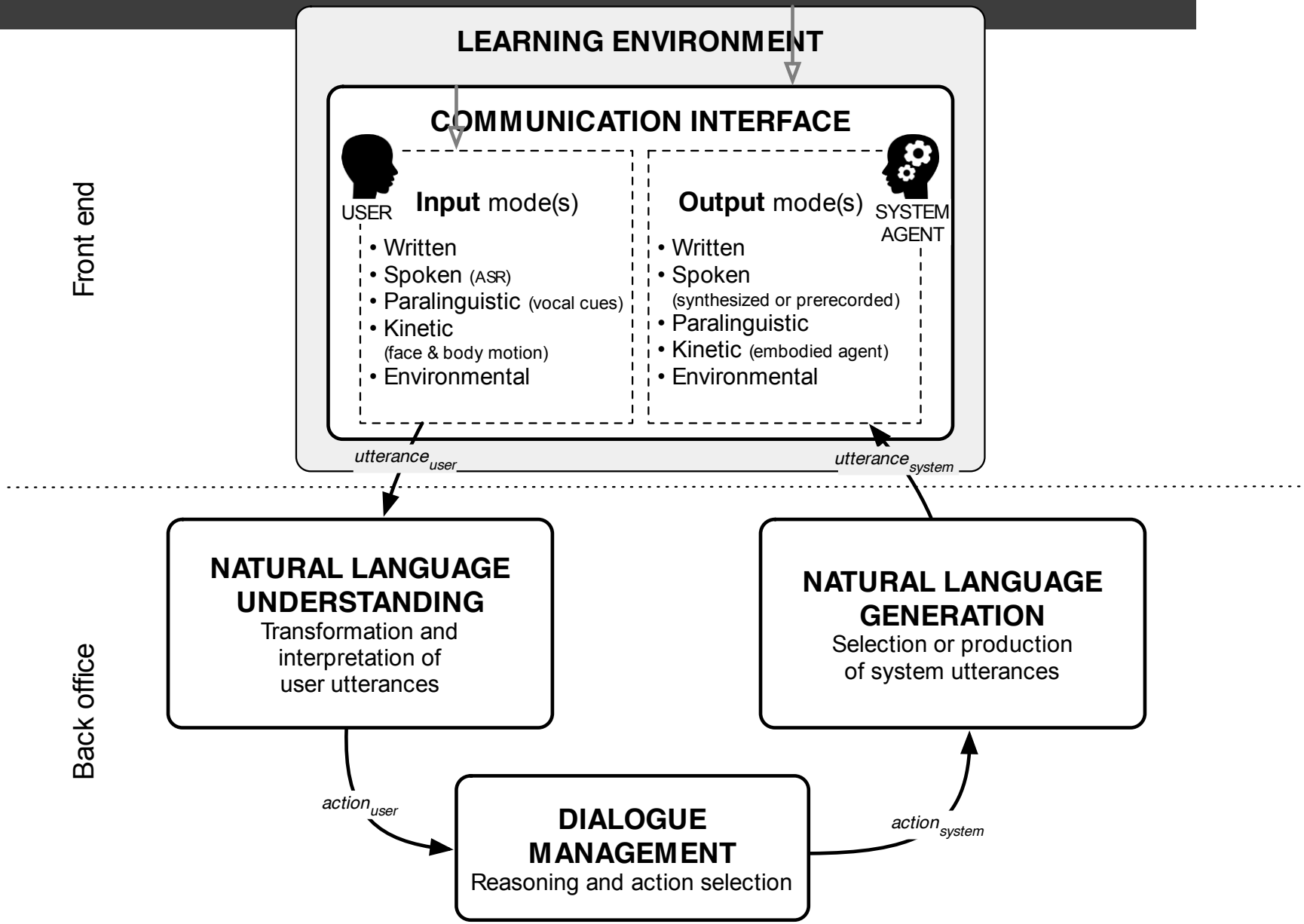
DBCALL: generic process



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Dialogue-based CALL: challenges and opportunities

- **What?**
  - Dialogue-based interaction with a computer, allowing negotiation of outcome through open learner output.
  - Interdisciplinary object (NLP/Speech, CALL, AI/HCI, Games/VW)
  - ⇒ **Need for more exchanges across research fields**
- **Why?**
  - Insufficient experimental results to demonstrate effectiveness for language learning
  - ⇒ **Need for more effectiveness research**
- **How?**
  - Advancements in dialogue systems have not yet been applied to DBCALL:
    - natural language understanding through dialogue act recognition
    - dialogue management with planning and conversational memory
    - data-driven & probabilistic models
  - ⇒ **Need for more technological research**

<User> Thank you for your attention!
<Mike> You are very welcome.
<User> Any questions?
<Mike> Not any more.
<User> Mike, let them speak.
<Mike> What is one luxury item you really want to have?
<User> Mike, that's enough.
<Mike> I agree.

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